

Data Cleaning Change Log

DecodeLabs Data Analytics Internship | Project 1: Data Cleaning & Preparation
Prepared by Muhammad Waqar Tahir

Dataset	E-commerce orders (data/raw/orders_raw.csv), 1,200 rows x 14 columns, orders dated 2023-01-01 to 2025-06-30
Rows in / out	1,200 in, 1,200 out (0 removed)
Changes applied	3 logged changes, 5 audit checks found nothing to fix
Verification gate	PASSED: 0 duplicate IDs, 0 incorrectly formatted dates
Raw SHA-256	ee0e0653974cd865373692f00469260f258865fb178b709e900d3fb534107033
Cleaned SHA-256	5450971f50695d544710cc50cc96392759b46c46ac4a1da5edaf5645d9d961e3

Headline finding. The raw file arrived in better shape than a typical extract: no duplicate records, no malformed dates, no arithmetic errors and no stray whitespace. The one substantive gap was 309 blank CouponCode values, which mean *no coupon was used* rather than *data was lost*, so they were labelled instead of deleted or guessed. The remaining changes are formatting only and alter no numeric value.

1. Change Log

Change ID	Description	Impact	Status
CR001	CouponCode blanks replaced with the explicit label 'No Coupon' Columns: CouponCode	309 records. Kept every affected row instead of deleting it; mode imputation rejected because it would fabricate coupon usage	Resolved
CR002	UnitPrice and TotalPrice fixed to exactly 2 decimal places; Quantity and ItemsInCart stored as integers Columns: UnitPrice, TotalPrice, Quantity, ItemsInCart	260 records. Cells reformatted: UnitPrice 114, TotalPrice 260 (e.g. 224 to 224.00). No value changed, only its representation	Resolved
CR003	Saved as UTF-8 CSV with LF line endings (raw file used Windows CRLF) Columns: file	1,200 records. Clean line-by-line diffs in Git; no cell values affected	Resolved

2. Audit Checks With Nothing To Fix

Each check below was implemented in the pipeline and run against the raw data. They are listed so a reviewer can see they were tested, not skipped.

Check ID	Check	Result	Status
AU001	Leading, trailing or repeated spaces in any column	0 records	No issue found
AU002	Missing values outside CouponCode	0 records	No issue found
AU003	Inconsistent capitalisation in text, ID and coupon fields	0 records	No issue found
AU004	Dates not in ISO 8601 (YYYY-MM-DD) format	0 records	No issue found
AU005	Duplicate rows or repeated OrderIDs	0 records	No issue found
AU006	Duplicate TrackingNumbers	0 records	No issue found
AU007	IDs not matching expected format (ORD#####, C#####, TRK#####)	0 records	No issue found
AU008	TotalPrice not equal to Quantity x UnitPrice	0 records	No issue found
AU009	Non-numeric entries in numeric columns	0 cells	No issue found

3. Verification Gate

Run against **orders_cleaned.csv as saved on disk** (re-read from the file, not from memory). The two gate checks are the Project 2 threshold from the training kit.

Check	Failures	Result
Duplicate OrderIDs (gate)	0	PASS
Incorrectly formatted dates (not YYYY-MM-DD) (gate)	0	PASS
Duplicate rows (all columns)	0	PASS
Duplicate TrackingNumbers	0	PASS
Missing / null-like values (all columns)	0	PASS
IDs not matching expected pattern	0	PASS
Prices not at 2 decimal places	0	PASS
Non-numeric Quantity / UnitPrice / ItemsInCart / TotalPrice	0	PASS
TotalPrice differs from Quantity x UnitPrice	0	PASS
Row count differs from expected	0	PASS

4. Data Observations (Flagged, Not Altered)

These are not cleaning errors. They are properties of the source data that anyone analysing it should know about. Nothing here was changed, because altering it would mean inventing data.

#	Observation
1	Coupons are not reflected in TotalPrice: it equals Quantity x UnitPrice on all 1,200 rows, so discount and shipping effects cannot be measured from this file.
2	497 orders (41.4%) are Cancelled or Returned but still carry a TotalPrice. Filter on OrderStatus before calculating revenue.
3	487 Cancelled or Pending orders have a TrackingNumber, which is unusual for orders that never shipped. Kept as supplied; confirm with the data owner.
4	UnitPrice varies widely inside a single product (Laptop: 18.20 to 699.93), so it is not a stable catalogue price. Kept as supplied.
5	8 TotalPrice values fall outside the 1.5 x IQR fence. They are valid high-quantity, high-price orders and were retained.
6	11 CustomerIDs appear on more than one order. These are repeat customers with different OrderIDs, not duplicates, and were retained.

5. Method and Reproducibility

Cleaning was done in Python (pandas) rather than Excel so the workflow is scripted and repeatable. The file is loaded as text so nothing is silently coerced, audited, cleaned in a fixed order (whitespace, missing values, case, dates, numeric types, duplicates), written to disk, then re-read and verified. The script exits with an error if any check fails. Automated tests exercise the same logic on a deliberately messy synthetic table.

```
python src/clean_data.py
python src/generate_change_log.py
python -m pytest tests
```